Adaptive Learning of Forest Graph States

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Abstract

Given a quantum state encoding an unknown labelled forest, how many copies does it take to recover every edge using single-qubit X/Z measurements? By using earlier copies to choose later measurements, a learner can repeatedly find vertices with at most two remaining neighbours. This recovers any n-vertex forest from $O((\log n)^2)$ copies with probability at least $2/3$. A fixed schedule requires $\Theta(n)$ copies in the worst case, even on connected trees.

1 Introduction

Suppose we know the labels of the vertices in a network and that its edges form a forest, an undirected graph without cycles. Which pairs of labels are joined? We receive independent preparations, or *copies*, of the same quantum graph state. Each copy has one qubit per vertex and encodes the entire edge set. The degrees and the number of components are unknown. Success means recovering every edge with its correct endpoints.

If we can hold two copies at once, we can measure the qubits carrying the same label jointly in the Bell basis [MS22]. This returns a random bit per vertex and tells us whether an even or odd number of its neighbours received a 1. These parities suffice to identify a forest using $O(\log n)$ copies at constant success probability (Appendix D). Suppose instead we measure each qubit separately in X or Z, choosing all bases on a copy before seeing its outcomes. A *fixed* learner chooses the whole schedule in advance, possibly at random. An *adaptive* learner can use earlier outcomes to choose the next settings. Both may keep every bit. *How many copies can we save by using that feedback?*

A forest can have a hub of degree $n-1$, yet at least a third of its vertices have at most two neighbours. The same fact holds in every remaining forest after vertices are removed. A batch of measurements can identify many short neighbour lists at once. After recording their edges, the learner knows which vertices to measure in Z on the next copy. Their bits help it test the next layer, even at a hub. A fixed learner must choose those settings before it knows which vertices they are.

Theorem 1 (Main result). *For exact recovery of any labelled n-vertex forest with probability at least $2/3$, adaptive single-qubit X/Z measurements suffice with $O((\log n)^2)$ copies. The optimal worst-case cost of fixed single-qubit X/Z measurements is $\Theta(n)$ copies. Both statements hold for connected trees as well.*

Proof. Theorem 4 gives the adaptive algorithm. Theorems 6 and 7 give the fixed lower and upper bounds, and Theorem 8 gives the lower bound for connected trees. □

Table 1: Worst-case copy guarantees at success probability $2/3$. Columns evaluate the adaptive budget (2) and the fixed lower bound $\lceil n/6 \rceil$ for even n.

Vertices n	Adaptive: at most	Fixed: at least
256	4,920	43
100,000	18,750	16,667
1,000,000	26,604	166,667
10,000,000	34,891	1,666,667

At one million vertices, 26,604 copies suffice for adaptive recovery, while every fixed learner needs at least 166,667 copies to give the same guarantee on every forest (Table 1). The first two rows leave the comparison undecided because the bounds overlap there. As n grows, the ratio of the fixed lower bound to the adaptive upper bound grows without bound.

Related work. Ouyang and Tomamichel [OT22] test proposed neighbour lists when every vertex has the same degree. Arunachalam et al. [ABDY23] use a single X measurement among Z measurements to read a derivative of a state's phase. For a graph state this derivative is the parity of the neighbouring bits. Our algorithm tests many lists on each copy and uses the forest structure to decide which vertices to test next. Montanaro and Shao [MS22] learn an unknown star in a constant number of copies. For our fixed-schedule lower bound, the difficult edge joins the hubs of two stars.

Gupta, He, and O'Donnell [GHO25] use feedback *within* a copy to certify a supplied target state. Goldar et al. [GQZ+26] use adaptation to estimate states from a constructed family. Here the target is an unknown labelled network, and all measurement choices based on feedback take place *between* copies.

Future directions. The optimal adaptive X/Z cost lies between $\Omega(\log n)$ and $O((\log n)^2)$ copies. Can single-qubit measurements close this gap? A second question allows controlled-NOT (CNOT) gates between qubits of the same copy, with each copy's circuit chosen using earlier outcomes. Could this achieve $O(\log n)$ copies for every forest? The Bell benchmark couples qubits from two different copies.

2 Reading an edge from measurement outcomes

On copy 12 of a recorded run, vertex 4 was measured in X and returned 0. The other five vertices were measured in Z. What can this row tell us about the neighbours of 4? Figure 1 shows the answer to the reader. The learner sees only the labels, settings and bits.

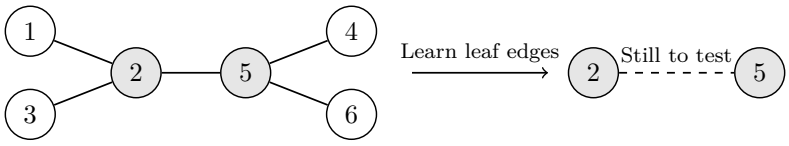


Figure 1: The hidden tree (left). Once the edges of leaves $1, 3, 4, 6$ have been learned, only the possible edge $\{2, 5\}$ remains to be tested (right). Every copy encodes the full tree.

The Z basis reads a computational-basis bit, 0 or 1. In the X basis the plus state gives 0 and the minus state gives 1. The list of bits from one copy is a *row*. For now the learner tests lists of at most two neighbours. Among five other labels there are 16 such lists: the empty list, five singletons and ten pairs. To test a list, count its Z bits equal to 1. An even count predicts 0 at vertex 4, and an odd count predicts 1. Consider just two rows from the same fixed schedule.

Vertex	1	2	3	**4**	5	6
Setting on both copies	Z	Z	Z	**X**	Z	Z
Bits on copy 12	1	0	1	**0**	0	0
Bits on copy 22	1	1	1	**1**	1	1

On copy 12, the empty list predicts 0, as do the singletons $\{2\},\{5\},\{6\}$. Four pairs also predict 0: the three pairs drawn from $2,5,6$ and the pair $\{1,3\}$, whose two ones give an even count. Eight lists survive. On copy 22, all five Z bits are 1 and vertex 4 returns 1. Only a singleton can give an odd count. Together the two rows leave exactly $\{2\},\{5\},\{6\}$. A disagreement rules out a list; several lists can agree on the same row.

The even-or-odd count is called a *parity*. Write b_v for the bit at vertex v, $N(v)$ for its neighbour set, and $\oplus$ for addition modulo two. Why can we trust a disagreement, and how often will a wrong guess disagree?

Lemma 2 (Testing a neighbour list). *On a fresh graph state, measure v in X and the vertices in a proposed list W, with $v \notin W$, in Z. The equality $b_v = \bigoplus_{u\in W} b_u$ holds with certainty when $W = N(v)$, and with probability $1/2$ otherwise. The empty parity is 0.*

Proof. Each qubit starts in $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. A controlled-$Z$ gate on edge $\{i,j\}$ changes an amplitude's sign when $x_i = x_j = 1$. If E is the edge set, the state is therefore

$$|G\rangle = 2^{-n/2} \sum_{x\in\{0,1\}^n} (-1)^{q_G(x)} |x\rangle, \qquad q_G(x) = \sum_{\{i,j\}\in E} x_i x_j \pmod 2. \tag{1}$$

First measure every other vertex in Z. Their bits are independent and fair because every string in (1) has the same squared amplitude. Compare the two amplitudes obtained by setting $x_v = 0$ and $x_v = 1$. Every neighbour with bit 1 changes their relative sign once. With an even count, the state at v is $|+\rangle$ up to an overall sign and its X bit is 0. With an odd count it is $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ up to sign and its bit is 1.

If W is wrong, its parity differs from the true parity by the sum of a nonempty set of independent fair bits. That sum is fair. Finally, the test reads only v and W. Summing over the outcomes at any other vertex gives the identity in every basis, so its setting leaves the test probabilities unchanged. □

On copy 22 the five Z ones have probability $1/32$. Given those ones, the true neighbour 5 fixes the X bit at 4 to 1. We will also use the operator form of this rule: $K_v = X_v Z_{N(v)}$ satisfies $K_v|G\rangle = |G\rangle$, where $Z_W = \prod_{u\in W} Z_u$. An operator that fixes the state in this way is called a *stabilizer*.

3 Using feedback to peel the forest

In a separate adaptive run on the same tree, the first batch of 128 copies identified the four leaves and their edges. Now vertex 2 has two known neighbours, 1 and 3, and only one possible further neighbour, 5. On copy 138 the settings in label order were $ZXZZZZ$ and the outcome was 001011. Subtract the contributions of the known neighbours by adding their bits modulo two:

$$b_2 \oplus b_1 \oplus b_3 = 0 \oplus 0 \oplus 1 = 1.$$

The empty remaining list predicts 0; the list $\{5\}$ predicts $b_5 = 1$. Given the learned leaf edges, this row settles the central edge. The implementation records it at the end of the batch. Earlier copies determined *which* bits to use; all three bits in the calculation came from copy 138.

Why try pairs as well as singletons? On a long path, removing only leaves strips off two endpoints per round. Every vertex already has at most two neighbours, so a successful round that tests pairs learns the whole path.

Algorithm 1. Let U contain the unresolved vertices. Initially U contains every label and the edge list is empty. For each $v \in U$, write B_v for its learned neighbours outside U. Keep these sets fixed within a round.

1. **Measure.** On each of L fresh copies measure every removed vertex in Z. At each unresolved vertex independently choose X or Z by a fair coin.

2. **Test.** For each $v \in U$, try every set T of at most two other unresolved vertices, including the empty set. Use a row when v is in X and $B_v \cup T$ is in Z. Eliminate T if the parity on $B_v \cup T$ disagrees with b_v. A list with no usable row survives.

3. **Peel.** Wherever exactly one list survives, record its edges. Then remove all these vertices together. Return the edge list when U is empty; report failure if vertices remain after R rounds. The one-vertex forest needs no copies.

How large must a batch be to find all vertices with at most two remaining neighbours, even though many guesses share the same rows?

Lemma 3 (Which vertices a batch peels). *Conditional on any correct history of earlier rounds, the probability that any wrong list survives L fresh copies is at most $n^3 e^{-L/16}$. If none survives, the algorithm peels exactly the vertices with at most two unresolved neighbours.*

Proof. All vertices in B_v are measured in Z. A row can test a particular T whenever at most three independent coin tosses take the required values: X at v and Z on T. This happens with probability at least $1/8$. A wrong list fails on half of its usable rows by Lemma 2. Independence across copies bounds its survival probability by $(1-1/16)^L \le e^{-L/16}$. Sum over the at most n^3 lists across all vertices. The true list always survives whenever it is tried, which occurs exactly when at most two unresolved neighbours remain. □

Theorem 4 (Adaptive recovery). *For every labelled forest on $n \ge 2$ vertices and every $0 < \delta < 1$, Algorithm 1 recovers all edges with probability at least $1-\delta$ using $O(\log n[\log n + \log(1/\delta)])$ copies.*

Proof. After each correct round the recorded edges are exactly the edges touching removed vertices. This holds initially. In the next correct round every newly peeled vertex has revealed all its remaining neighbours. Recording edges before removing vertices captures even an edge whose endpoints leave together. This proves the claim by induction.

Number of rounds. A forest on $m > 0$ remaining vertices has fewer than m edges, so its degree sum is below $2m$. Fewer than $2m/3$ vertices can have degree at least three. A correct round leaves only these vertices. Removing at least a third each time takes $O(\log n)$ rounds.

Copies per round. Choose the round cap R and batch size L as

$$R = \lceil \log_{3/2} n \rceil + 1, \qquad L = \lceil 16 \ln(Rn^3/\delta) \rceil. \tag{2}$$

Lemma 3 makes the conditional probability of a first mistake in any round at most δ/R. Adding over R rounds gives failure probability at most δ. At fixed confidence $L = O(\log n)$: each wrong list's chance of survival decreases exponentially, and there are only polynomially many lists. Multiplying the rounds by the copies per round gives the two logarithms. □

The decoder can compute the parity on B_v once per vertex and row. Each candidate then uses at most two additional bits, for $O(n^3 RL)$ elementary operations and $O(n^3)$ candidate flags.

4 What fixed measurements cost

Deleting edge $\{2, 5\}$ from the six-vertex tree leaves two stars. Which rows reveal the bridge in Figure 2? Once the Z bits of hubs 2 and 5 are fixed, adding their edge changes every remaining amplitude by the same sign, negative only if both bits are 1. The outcome probabilities agree. A private leaf can fix its hub's bit.

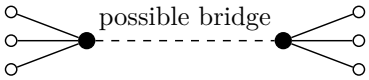


Figure 2: A fair coin adds or omits the bridge between two stars.

Lemma 5 (When a bridge is invisible). *Suppose two graphs differ only in edges between hubs. For a fixed X/Z setting, each hub is in Z or has a common private leaf in X, outside the hub set and adjacent only to that hub. Then every outcome string has the same probability in both graphs.*

Proof. Read one such outcome per hub first. A Z reading fixes the hub bit directly. A private leaf ℓ in X fixes its hub's Z bit since $X_\ell Z_v$ stabilizes the state. Switching the hub edges applies controlled-Z gates that commute with these readings. On the projected state the gates contribute one overall sign. The squared norm and all subsequent outcome probabilities agree. □

Randomly hiding the two hubs makes rows that violate the lemma's condition rare.

Theorem 6 (A hidden bridge forces a linear cost). *Every fixed X/Z learner that recovers each labelled forest with probability at least $2/3$ requires at least $n/6$ copies in the worst case for even $n \geq 4$, and at least $(n-1)/6$ for odd $n \geq 5$.*

Proof. Randomly split the labels into equal blocks of size $s = n/2$. Choose a uniform hub independently in each block, attach its private leaves and flip a fair coin for the bridge. A row *exposes* a hub if it is its block's only X.

Fix a row with k labels in X. When $k = 0$, exposure is impossible. For $k = 1$ or $k \geq 3$, at most one block contains a single X, which hits its hub with probability $1/s$. For $k = 2$, the labels fall in separate blocks with probability $s/(2s-1)$. Given this split, at least one hits its hub with probability $1 - (1 - 1/s)^2$. The product is again $1/s$.

Fix the schedule's random seed and pad early stops to t rows. The chance of any exposure is at most t/s. Conditional on roles and settings with no exposure, Lemma 5 gives identical row laws for both bridge choices. Independent copies then give identical laws for the *entire record*. Every decoder guesses the fair coin with success $1/2$ on this event. Even perfect decisions elsewhere give success at most $1/2 + t/(2s) = 1/2 + t/n$ after averaging over seeds. Counting the recovered edges reveals the coin, so $t \geq n/6$. For odd n, add a common isolated vertex. □

4.1 A matching fixed schedule

Late in peeling, testing one vertex may require many known neighbours to be in Z together. A fixed schedule can provide such rows by making X progressively rarer.

Theorem 7 (A matching fixed schedule). *For $n \geq 2$ and $0 < \delta \leq 1/3$, a randomized fixed X/Z schedule recovers every forest with probability at least $1 - \delta$ using $O(n[1 + \log(1/\delta)])$ copies.*

Proof. Preassign four batches for each $M = 2^{\lceil \log_2 n \rceil}, \ldots, 2, 1$, in that order. Each batch has L_M independent rows. In every row choose X independently at each label with probability x, where

$$\Delta = \lceil 24n/M \rceil, \qquad x = 1/(\Delta + 3), \qquad L_M = \lceil 2e(\Delta + 3) \ln(8M^4/\delta) \rceil.$$

Starting with at most M unresolved vertices, aim to leave at most $M/2$. The decoder then ignores further batches at this scale. All four count toward the budget.

In each batch test vertices with at most Δ known neighbours, trying each set of up to two other unresolved vertices. Given a correct history, a row is usable for a guess with probability at least $x(1-x)^{\Delta+2} \geq 1/[e(\Delta+3)]$. A wrong guess fails on half its usable rows. Summing over at most M^3 guesses gives probability at most $M^3 e^{-L_M/[2e(\Delta+3)]} \leq \delta/(8M)$ that any wrong one survives.

If $m > M/2$ vertices remain, more than $m/3$ have at most two unresolved neighbours. At most $2n/\Delta \leq M/12 < m/6$ vertices in the original forest have degree above Δ. Thus at least $m/6$ vertices qualify and have a true guess. When all wrong guesses fail, record the edges of those vertices and remove them. Four correct batches reach the goal since $(5/6)^4 < 1/2$. The last scale leaves zero vertices. Summing the probabilities of a first error gives at most $4 \sum_M \delta/(8M) < \delta$. At $M = 2^j$, a batch costs $O(n2^{-j}[1 + j + \log(1/\delta)])$ copies. Summing over j proves the bound. □

Thus the fixed-schedule cost is $\Theta(n)$ at fixed confidence. Appendix A proves the linear lower bound even for connected trees.

5 Recovery at a fixed budget and as the budget grows

Each trial asked whether a method recovered every edge of a sampled network. To draw one outcome, the simulator generates a fresh fair bit for each vertex. A vertex in Z reports its own bit, and a vertex in X reports the parity of its neighbours' bits. Only the simulator knows the graph. Lemma 12 proves that these rows have the same joint distribution as quantum measurements of a forest.

We compare adaptive peeling with three fixed schedules. *Fair* chooses X or Z independently with equal probability at every vertex. *Mixed* combines fair rows with rows containing a single X. *Multiscale* varies the probability of X over $1/2, 1/4, \ldots, 2^{-\lceil \log_2 n \rceil}$, with a different allocation from Theorem 7. All three fixed decoders can reuse earlier rows after learning an edge. Adaptive peeling uses fresh batches, with sizes selected on 24 separate calibration trials per network shape and size.

Table 2: Exact recoveries from 512 copies on 256 vertices, out of 128 noiseless trials per network shape. Bold marks a sole leader.

		Fixed schedule		
Network	Adaptive	Fair	Mixed	Multiscale
Randomly labelled path	128	128	128	128
Uniform labelled tree	1	0	0	**83**
Tree with edges deleted	87	33	0	**111**
Two stars	**126**	0	89	53

What do 512 copies buy? At 256 vertices every method recovered all 128 paths (Table 2). Fixed multiscale did best on uniform labelled trees, with 83 recoveries against one for adaptive

peeling. It also led 111 to 87 when each tree edge was retained independently with probability 0.7. Adaptive peeling did best on two stars with a bridge chosen by a fair coin, recovering 126 against 89 for fixed mixed. At this budget the adaptive method has just two batches of 256. Its worst-case guarantee at this size allows 15 rounds of 328 copies.

What changes with more copies? On the six-vertex example, fixed mixed completed recovery at 128 copies and adaptive peeling at 256. Across the four noiseless shapes at $n = 32, 64, 128, 256$, with 128 held-out trials per shape and size, adaptive peeling never reached two-thirds recovery at an earlier tested budget than the best fixed method selected in hindsight. The two-star curves at $n = 256$ show how this can coexist with a higher adaptive recovery count: adaptive and fixed mixed first crossed two thirds at the same 512-copy budget (Figure 3, left).

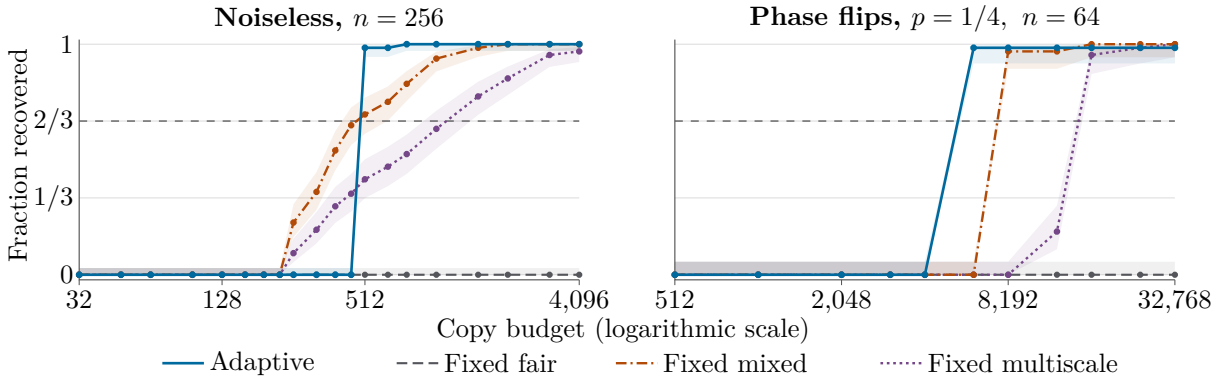


Figure 3: Exact recovery of two-star networks as the copy budget increases. Each point gives the fraction recovered in 128 noiseless trials (left) or 64 noisy trials (right). Lines join tested budgets. Shading gives pointwise 95% Wilson intervals. The horizontal dashed line marks two-thirds recovery.

In the noisy trials each qubit suffered an independent phase flip with probability 1/4. The study used 64 trials per shape at $n = 32, 64$. On 64-vertex two-star networks, adaptive peeling first exceeded two thirds at 6,144 copies, recovering 63 of 64. Fixed mixed recovered none at 6,144 and 62 at 8,192 (Figure 3, right). Here adaptive batches had 3,072 copies and each candidate needed 64 usable rows (Appendix C).

These comparisons depend on the shape, budget and decoder. The asymptotic advantage in Table 1 concerns the copies needed to guarantee recovery on *every* forest. The study recorded root seed 20261004. The six-vertex sampler agrees with the quantum probabilities for all 4,096 setting–outcome combinations. The two-star curves preserve the archived aggregate counts; the trial-level records and runnable study code were unavailable for an independent rerun.

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A Four hubs in a connected tree

Two trees on the same labels have the same edge count. To hide their difference, change the order of four hubs on a path.

Theorem 8 (A hidden path in a connected tree). *For $n \geq 8$, every fixed X/Z learner recovering each connected labelled tree with probability at least $2/3$ needs at least $\lfloor n/4 \rfloor/12$ copies in the worst case. For $n = 4s$, the bound holds even given the four blocks below.*

Proof. Partition $4s$ labels into four blocks of size s. In each block choose a uniform hub independently and attach the other labels as leaves. Independently flip a fair coin to connect the four hubs a, b, c, d by the path a, b, c, d or a, c, b, d. In one fixed row the chance of exposing a given hub is at most $1/s$. In t rows the chance of any exposure is at most $4t/s$. Otherwise the entire record has the same distribution for both paths by Lemma 5. The recovered tree reveals the path, so $1/2 + 2t/s \geq 2/3$, giving $t \geq s/12$. For $n = 4s + r$, attach the $r < 4$ extra labels as leaves of a. The original blocks still supply all private leaves needed by the lemma. □

Why 48 adaptive copies suffice for this family. Suppose $n = 4s$ with $s \geq 3$, and the partition is hidden. Measure eight copies entirely in X and group labels by their eight-bit strings. By Lemma 12, every private leaf returns the hidden fair bit of its hub. Conditional on all four hub bits, each hub's output contains an independent fair parity from its own leaves. Thus the four hub strings and the four leaf-group strings are independent and uniform. A collision has probability at most $\binom{8}{2}2^{-8}$. With no collision, the four singleton groups identify the hubs since every leaf group has size at least two.

For each hub, try all 32 lists comprising one of the four leaf groups and any subset of the other three hubs. Measure the chosen hub in X and all other labels in Z on ten fresh copies. Lemma 2 gives each wrong list survival probability 2^{-10}. The total error is at most $\binom{8}{2}2^{-8} + 4 \cdot 31\, 2^{-10} < 1/3$, at a cost of $8 + 4 \cdot 10 = 48$ copies.

B What a third measurement can see

Our X/Z parity test at a hub uses a row only when all its known neighbours were measured in Z. A third basis, the eigenbasis of $Y = iXZ$, lets the decoder use some of the rows where this fails.

Theorem 9 (Fixed recovery with three bases). *For $n \geq 2$ and $0 < \delta \leq 1/3$, a randomized fixed schedule of single-qubit X, Y, Z measurements recovers every labelled forest from $O(\sqrt{n}[1 + \log(1/\delta)])$ noiseless copies with success probability at least $1 - \delta$. At success probability $2/3$, every such learner needs $\Omega(\sqrt{n})$ copies in the worst case, even on connected trees.*

Proof. A fixed schedule. Use the four-batch scales of Theorem 7. At scale M, put $\Delta = \lceil 48n/M \rceil$. For each unresolved vertex v, count the two-edge paths from v through its known neighbours B_v by $H_v = \sum_{d \in B_v} (\deg(d) - 1)$. Each removed d has at most two neighbours still unresolved. Hence $\sum_v H_v \leq 2 \sum_d \deg(d) \leq 4n$. Skip vertices with $H_v > \Delta$, at most $M/12$ of them.

Fix a guess T of at most two remaining neighbours. Preassign two blocks per batch, with independent bases at each site and row. If $|B_v| \leq \sqrt{\Delta}$, use the first block, which chooses X with probability $1/(\sqrt{\Delta} + 3)$ and Z otherwise. The ordinary test is usable with probability $\Omega(1/\sqrt{\Delta})$ as in Theorem 7.

If $|B_v| > \sqrt{\Delta}$, use the second block, which chooses Y with probability $q = 1/(32\sqrt{\Delta})$ and Z otherwise. Require v in Y and T in Z. Let J be the known neighbours of v in Y. If $|J|$ is odd and all other neighbours of each $d \in J$, apart from v, are in Z, read the signed product

$$P = (X_v Z_{B_v \cup T}) \prod_{d \in J} K_d, \qquad K_d = X_d Z_{N(d)}.$$

The decoder knows these K_d. Each fixes the state and commutes with the test through two sign changes, one at v and one at d. Multiplication gives Y at v and J, only Z or identity elsewhere, and sign $(-1)^{(|J|-1)/2}$, since neighbours of v are nonadjacent. Thus every factor is measured and the test has its original mean.

Given the required bases at v, T, the chance of odd $|J|$ is $[1 - (1-2q)^{|B_v|}]/2 \geq (1 - e^{-1/16})/2$. A forbidden pair of Y's on one of the H_v paths costs at most $H_v q^2 \leq 1/1024$ in total. The remaining conditional probability is a positive constant. Multiplying by $q(1-q)^{|T|}$ gives usability $\Omega(1/\sqrt{\Delta})$ in this block too.

A wrong guess fails on half its usable rows. Thus $C\sqrt{\Delta} \log(8M^4/\delta)$ rows per block, rounded up for sufficiently large C, give error at most $\delta/(8M)$ over at most M^3 guesses. While $m > M/2$ vertices remain, more than $m/3$ have degree at most two and at most $M/12$ were skipped. Four correct batches therefore halve M, as before. Summing the errors gives at most δ. Summing the copies gives the stated upper bound since $\sum_{j \geq 0} (1 + j) 2^{-j/2} < \infty$.

Why fewer copies cannot suffice. Make four public blocks of $N = 3k + 1$ labels. In each block randomly assign a hub v, k private leaves and k branches $v - u - w$. A fair bit joins the four hubs by path $a - b - c - d$ or $a - c - b - d$. Use the largest such blocks with $4N \leq n$, attaching leftover labels as leaves of a in both trees.

A Z reading at a hub reveals its bit directly. An X reading at a private leaf ℓ does the same, since $X_\ell Z_v$ fixes the state. Two Y readings on a branch also reveal it: $K_u K_w = Z_v Y_u Y_w$ fixes the state, so the product of the two signs gives the hub's Z sign. If all hubs have a witness, read these disjoint witnesses first. The hub bits are then fixed, and switching the path only changes an overall sign, as in Lemma 5. The full row distributions agree.

Fix a setting with x sites in X and y in Y in one block. The probability of an unwitnessed X hub is at most $(x/N)(2/3)^{x-1} = O(1/N)$ for $x \geq 1$, since the other X's must avoid all private leaves. For $y \leq \sqrt{N}$, a Y hub already has probability at most $1/\sqrt{N}$. Otherwise, conditional on a Y hub, the number L of branches with both endpoints u, w in Y has mean $(y-1)(y-2)/[3(N-2)]$, by counting the k disjoint pairs. Their indicators have nonpositive covariance because filling one pair leaves fewer Y's for the next. Thus $\mathrm{Var}(L) \leq \mathbb{E}L$ and $\Pr[L = 0] \leq 1/\mathbb{E}L$. Including the probability y/N of a Y hub gives $O(1/y) = O(1/\sqrt{N})$ for large N.

Over four hubs and t fixed rows the probability of any missing witness is $O(t/\sqrt{n})$. Otherwise the entire record has the same law for both paths. Averaging over roles and schedule gives the same upper bound on the total variation distance $\mathrm{TV}(P,Q) = \frac{1}{2}\sum_y |P(y) - Q(y)|$ between record laws. The best test of two equally likely laws chooses the more likely one for each record and succeeds with probability $(1 + \mathrm{TV}(P,Q))/2$. Success $2/3$ therefore requires $\mathrm{TV}(P,Q) \geq 1/3$, so $t = \Omega(\sqrt{n})$. □

C Phase errors and the classical simulation

Suppose each qubit on each copy independently receives a phase flip Z with known probability $0 \leq p < 1/2$. This reverses an X bit and leaves a Z bit alone. A single disagreement can now occur even for the correct neighbour list. Instead, we count how often a list passes.

Theorem 10 (Recovery under phase flips). *Let $n \geq 2$, $0 < \delta \leq 1/3$, and $\lambda = 1 - 2p$. Under these errors, every labelled forest can be recovered with probability at least $1 - \delta$ using $O(\lambda^{-2} \log n[\log n + \log(1/\delta)])$ adaptive X/Z copies or $O(\lambda^{-2} n[1 + \log(1/\delta)])$ fixed X/Z copies.*

Proof. On a correct history, score a pass as $+1$ and a failure as -1. Only the flip at the tested vertex changes the score. On every usable row its mean is therefore λ for a correct list and zero for a wrong one, by Lemma 2. Accept if at least one row is usable and the average score exceeds $\lambda/2$.

Suppose each of L independent rows is usable with probability r. Given the K usable settings, the scores are independent. Hoeffding's inequality bounds either decision error by $e^{-K\lambda^2/8}$, including the trivial bound one at $K = 0$. Averaging over $K \sim \mathrm{Bin}(L, r)$ gives

$$\Pr[\text{wrong decision}] \leq [1 - r(1 - e^{-\lambda^2/8})]^L \leq e^{-rL\lambda^2/16}. \tag{3}$$

The last step uses $1 - e^{-x} \geq x/2$ for $0 \leq x \leq 1$. If the implementation requires 64 usable rows, add $\Pr[K < 64]$ to the true-list rejection bound.

In adaptive peeling $r \geq 1/8$. Taking $L = \lceil 128\lambda^{-2} \ln(Rn^3/\delta) \rceil$ bounds the chance of any wrong decision among n^3 guesses by δ/R. For the fixed schedule at scale M, $r \geq 1/[e(\Delta + 3)]$, where $\Delta = \lceil 24n/M \rceil$. Taking $L = \lceil 16e(\Delta + 3)\lambda^{-2} \ln(8M^4/\delta) \rceil$ bounds the corresponding chance among M^3 guesses by $\delta/(8M)$. With correct decisions both algorithms peel exactly as before. Summing over the first erroneous batch as in Theorems 4 and 7 gives the claims. □

How much does phase noise cost a fixed learner? Even after it happens to expose a bridge, the measured hub bits can be flipped.

Theorem 11 (The fixed cost of phase flips). *In the same noise model, for $n \geq 4$ and $0 < \delta \leq 1/3$, any fixed X/Z schedule that recovers every labelled forest with probability at least $1 - \delta$ needs $\Omega(n\lambda^{-2} \log(1/\delta))$ copies in the worst case.*

Proof. In each of two known blocks of $s = \lfloor n/2 \rfloor$ labels, choose a uniform hub and join it to its block's other vertices. A fair bit adds or omits the bridge between hubs. Any spare vertex is isolated. Recovering the edge count decides the bit. A row exposes a hub only if that hub is its block's sole X label, so an exposure has probability at most $2/s$. Without exposure the two row distributions agree by Lemma 5, even after noise.

For distributions P, Q, their affinity $F = \sum_y \sqrt{P(y)Q(y)}$ measures overlap and equals one for identical laws. Fix a row and the hubs. Discard all X-measured hub bits. The remaining outcomes

have a common distribution because their measurement operators act on each hub as identity or a Z projector, both commuting with the bridge gate. The phase errors commute with it too.

Given the common outcomes, at most two ideal hub bits pass through independent flips. For two ideal strings differing in $j \le 2$ positions the output affinity is $(2\sqrt{p(1-p)})^j \ge 1-\lambda^2$, with value one at $j = 0$. Joint concavity extends this bound to mixtures of ideal strings. Also the noiseless states have overlap $1/2$, since the bridge changes the sign on one quarter of basis strings. For any noisy measurement with effects E_y, Cauchy–Schwarz gives $|\langle\psi_0|E_y|\psi_1\rangle| \le \sqrt{P(y)Q(y)}$. Summing gives $F \ge 1/2$. Thus an exposed row has $F \ge \max(1/2, 1-\lambda^2) \ge e^{-2\ln(2)\lambda^2}$.

Reveal the hubs and random schedule after all t rows (pad early stops). Conditional affinities multiply, and the expected number of exposures is at most $2t/s$. Averaging and applying Jensen's inequality gives $F_{\text{record}} \ge e^{-4\ln(2)\lambda^2 t/s}$. Error at most δ requires $1-2\delta \le \mathrm{TV}(P,Q) \le \sqrt{1-F^2_{\text{record}}}$. Hence $t \ge s\ln[1/(4\delta(1-\delta))]/[8\ln(2)\lambda^2]$, which gives the claimed order throughout the stated range. □

What distribution should the classical simulator draw to reproduce a whole quantum measurement row? Write $\mathbb{F}_2 = \{0,1\}$ with arithmetic modulo two. For a bit mask c, X^c applies X at its selected sites, and Z^c is defined likewise.

Lemma 12 (Simulating a forest exactly). *Let A be a forest's adjacency matrix over $\mathbb{F}_2$, and draw a fresh uniform $w \in \mathbb{F}_2^n$ per copy. Output w_i at each Z site and $(Aw)_i$ at each X site. This gives the exact joint measurement law. To include phase errors, independently flip each reported X bit with probability p.*

Proof. It suffices to match the mean sign of every parity, since these means determine a distribution on bit strings. For one parity let c, d indicate its X, Z sites. The simulated sign is $(-1)^{(Ac+d)\cdot w}$, whose mean is one if $Ac = d$ and zero otherwise. For the graph state, X^c shifts x to $x + c$. Using $q_G(x+c) + q_G(x) = q_G(c) + (Ac)\cdot x$ in (1) gives quantum mean

$$\langle G|X^c Z^d|G\rangle = (-1)^{q_G(c)} 2^{-n} \sum_x (-1)^{(Ac+d)\cdot x}.$$

When $Ac = d$, the disjoint supports of c, d imply even degree at every vertex of the forest induced by c. Any component containing an edge would have a leaf. Thus the induced forest has no edges and $q_G(c) = 0$. The means agree in both cases. Independent Z errors flip exactly the reported X bits. □

D How much could entangling measurements save?

Even a learner with unrestricted measurements needs enough copies to tell apart all labelled trees.

Theorem 13 (A lower bound for any measurement). *If a learner identifies every labelled tree on $n \ge 2$ vertices with probability at least $2/3$ using t copies, then $nt \ge (n-2)\log_2 n + \log_2(2/3)$, even with a joint measurement on all copies.*

Proof. There are n^{n-2} trees. Repeatedly delete the smallest leaf and write down its neighbour until two labels remain. To invert the word, join the smallest remaining label absent from the word to its first entry, delete that label and entry, and repeat. Finally join the last two labels. This pairs the trees with the words of length $n-2$ over n labels.

Choose a tree uniformly. Let E_G be the measurement effect for reporting tree G, and ρ_G its t-copy state. The success probability on G is $\mathrm{tr}(E_G\rho_G) \le \mathrm{tr}(E_G)$, since $\rho_G \le I$. All tree effects together sum to at most I, of trace 2^{nt}. Average success is therefore at most $2^{nt}/n^{n-2}$. It must be at least $2/3$, giving the bound. □

A matching benchmark using pairs of copies. Measure corresponding qubits of two preparations together in the Bell basis [MS22]. At each label this reads both $Z \otimes Z$ and $X \otimes X$. Collect their bits into vectors w and s, respectively. The vector w is uniform because it is the bitwise sum modulo two of two uniform computational-basis strings from independent copies. Since $K_v \otimes K_v$ fixes the pair, s_v is the parity of w on the neighbours of v. Thus $s = Aw$.

Repeat on t independent pairs. Each label now has a string of t independent w bits. Its s string is the sum of its neighbours' strings. Look for a zero sum (an isolated vertex) or a match with one other string (a leaf). Remove one matching vertex in label order and subtract its string from its neighbour's sum, if any. Every nonempty forest has a leaf or an isolated vertex. With correct tests this rule peels the entire forest.

Fix the order followed with correct tests. A wrong equality along this order would require a nonempty set of independent uniform strings to sum to zero, with probability 2^{-t}. There are at most n^3 comparisons on the order, and the actual decoder follows it until its first error. With $t = \lceil \log_2(n^3/\delta) \rceil$ pairs, the error is at most δ. The cost is $2t$ preparations. This benchmark uses gates between the two copies in a pair. The CNOT question in the introduction permits gates only inside one copy.